

Sowjanya SK Susarla

B.E. Computer Science & Engineering | CBIT, Hyderabad | CGPA 8.72/10

kameshwarisowjanya@gmail.com | +91-7989218633 | Hyderabad, India | github.com/Sowjanya12125 | linkedin.com/in/sowjanya-s-k-susarla

Summary

CS Engineering student (CGPA 8.72) with strong DSA/OOP/DBMS fundamentals and hands-on experience shipping full-stack, ML, and security systems end-to-end. Built a credit-risk classifier, an AI-powered assistant, and led enterprise security assessments at India Post – comfortable owning the full SDLC under tight deadlines.

Technical Skills

Languages: Python, SQL, JavaScript/TypeScript

CS Core: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Web/Backend: FastAPI, Next.js (App Router), React, REST API Design, Tailwind CSS

Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, SHAP, TensorFlow, MongoDB, Prompt Engineering

Security: Threat Modelling, Risk Pattern Analysis, Anomaly Detection, Probabilistic Risk Frameworks

Experience

Research Intern – Security & Risk Analytics

Jul 2025

India Post, CEPT, Hyderabad

- Ran an end-to-end security posture assessment on an enterprise-scale networked system, correlating risk signals across 5+ domains to surface non-obvious threat patterns.
- Built data integration pipelines stitching together incomplete, real-time threat feeds, cutting manual analyst triage effort and improving detection latency.
- Authored a technical brief on probabilistic risk models after identifying limits of rule-based detection; formally adopted by the team as a reference for ML-driven risk modelling.
- Delivered findings to 15+ senior non-technical stakeholders, translating statistical risk insights into actionable recommendations.

Projects

Jarvis – AI-Powered Personal Assistant

2026

Python, FastAPI, React, Vite, Gemini API, Tailwind CSS

- Built a full-stack, context-aware AI assistant with 7 modules (briefings, planning, deadline tracking, reply drafting, budgeting) on a FastAPI backend and React/Vite frontend.
- Engineered tone-matched reply generation across 6+ sender archetypes with multi-turn conversation memory via the Gemini API.
- Designed modular prompt-routing architecture mapping each module to a dedicated FastAPI route; deployed independently on Vercel and Render with Git-triggered CI/CD.

ExpenseFlow – Full-Stack Expense Tracker Dashboard

2026

Next.js 14, TypeScript, Tailwind CSS, Recharts

- Built a full-stack expense tracker exposing a complete typed REST API (GET/POST/PUT/DELETE) with server-side validation shared across API routes, data layer, and UI.
- Developed a live analytics dashboard with instant-updating stats, a category-wise donut chart, and a 6-month spending trend via Recharts.
- Architected a repository-pattern data layer enabling a one-file swap to Prisma/PostgreSQL without rewriting application logic.

Credit Risk & Loan Default Prediction System

2025

Python, Scikit-learn, Pandas, Logistic Regression, Random Forest, XGBoost, SHAP

- Designed and deployed a modular, OOP binary classification pipeline for loan default prediction, achieving 87% accuracy on 50K+ held-out records.
- Engineered 12+ domain-specific features (debt-to-income ratios, payment history trends), boosting F1-score by 9%.
- Benchmarked Logistic Regression, Random Forest, and XGBoost with cross-validation and SHAP-based explainability for auditable risk scores.

Hackathons & Achievements

- **Round 2 – ISRO Bharatiya Antariksh Hackathon (National), Jul 2025** – Advanced among hundreds of nationwide teams; built high-complexity data processing algorithms for large-scale satellite and aerospace datasets, presented to ISRO evaluators.

Education

Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad

2023 – 2027

B.E. Computer Science & Engineering – CGPA: 8.72/10.0

Certifications

Cisco Networking Essentials – Cisco (Dec 2025) | Ethereum & Solidity (Blockchain Development) (Oct 2025)